

Chapter ① Numbers

I : Numeral :- In Hindu Arabic system, we use ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 called digits to represent any number. A group of digits, denoting a number is called a numeral.

e.g. 689745132

6	Ten crores	10^8
8	crore	10^7
9	Ten lacs (Million)	10^6
.		
7	Lacs	10^5
.		
4	Ten thousands	10^4
5	Thousands	10^3
1	Hundreds	10^2
.		
3	Tens	10^1
.		
2	Units	10^0

Sixty-eight crores, ninety seven lacs, forty five thousand, one hundred and thirty two.

II : Place value or local value of a digit in a numeral :

In the above numeral :

Place value of 2 is $2 \times 1 = 2$

place value of 3 is $3 \times 10 = 30$

place value of 1 is $1 \times 100 = 100$

place value of 6 is $6 \times 10^8 = 600000000$.

III: Face value: The face value of a digit in a numeral is the value of the digit itself at whatever place it may be.

Face value of 2 is 2.

Face value of 3 is 3.

IV: Types of numbers:

1. **Natural numbers:** (counting numbers)

$$N = 1, 2, 3, \dots$$

2. **Whole numbers:** All counting numbers together with zero form the set of whole numbers. Thus,

* 0 is the only whole number which is not a natural number.

* Every natural number is a whole number.

3. **Integers:** All natural numbers, 0 and negative of counting numbers, i.e.

$\dots - 3, -2, -1, 0, 1, 2, 3, \dots$ together form the set of integers.

* positive integers: $\{1, 2, 3, 4, \dots\}$ is the set of all positive integers.

* Negative integers: $\{-1, -2, -3, \dots\}$ is the set of all negative integers.

* Non-positive and non-negative integers:

0 is neither positive nor negative. So

$\{0, 1, 2, 3, \dots\}$ represents the set of non-negative integers, while $\{0, -1, -2, -3, \dots\}$ represents the set of non-positive integers. (2)

4. Even numbers:

5. Odd numbers:

6. Prime numbers: A number greater than 1 is called a prime number, if it has exactly two factors, namely 1 and the number itself.

Prime numbers upto 100: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Prime numbers greater than 100: let p be a given number greater than 100. To find out whether it is a prime or not, we use the following method:

Find a whole number nearly greater than the square root of p . let $k > \sqrt{p}$. Test whether p is divisible by any prime number less than k . If yes, then p is not prime, otherwise, p is prime.

e.g. we have to find whether 191 is a prime number or not. Now $14 > \sqrt{191}$.

Prime numbers less than 14 are

2, 3, 5, 7, 11, 13.

191 is not divisible by any of them. So 191 is a prime number.

7. Composite numbers:- Numbers greater than 1, which are not prime, are known as composite

numbers, e.g. 4, 6, 8, 9, 10, 12, ...

* 1 is neither prime nor composite.

* 2 is the only even number which is prime.

* There are 25 prime numbers between 1 and 100. **Co-primes**:- Two numbers a and b are co-primes if their HCF is 1. e.g. (2, 3), (4, 5), (7, 9), (8, 11)...

V: Tests of divisibility :-

1. **Divisibility by 2**: A number is divisible by 2, if its unit's digit is any of 0, 2, 4, 6, 8.

Ex- 84932 ✓
65935 X

2. **Divisibility by 3**:- A number is divisible by 3, if the sum of its digits is divisible by 3.

e.g. 592482 $\rightarrow 5+9+2+4+8+2 = 30 = 3 \times 10$
 $= 3$ ✓

864329 $\rightarrow 8+6+4+3+2+9 = 32 = 3+2 = 5$ X

3. **Divisibility by 4**:- A number is divisible by 4, if the number formed by the last two digits is divisible by 4.

e.g. 892648 $\rightarrow 48$ is divisible by 4 ✓

e.g. 749282 $\rightarrow 82$ is not divisible by 4 X

4. **Divisibility by 5**: A number is divisible by 5, if its unit's digit is either 0 or 5.

5. **Divisibility by 6**:- A number is divisible by 6, if it is divisible by both 2 and 3.

e.g. 35256 ✓ $\sqrt{798} \rightarrow \begin{array}{r} 8 \times 2 = 16 \\ 79 - 16 = 63 \end{array}$ is divisible

6. Divisibility by 8 :- A number is 3 divisible by 8, if the number formed by the last three digits of the given number is divisible by 8.

e.g. 953360 $\rightarrow$ 360 is divisible by 8. ✓

e.g. 529418 $\rightarrow$ 418 is not divisible by 8 ✗

7. Divisibility by 9 :- A number is divisible by 9, if the sum of its digits is divisible by 9.

e.g. 60732 $\rightarrow 6+0+7+3+2 = 18$ is divisible by 9 ✓

e.g. 68956 $\rightarrow 6+8+9+5+6 = 34$ is not divisible by 9. ✗

8. Divisibility by 10 :- A number is divisible by 10 if it ends with 0.

e.g. 96410, 10480

9. Divisibility by 11 :- A number is divisible by 11, if the difference of the sum of its digits at odd places and the sum of its digits at even places, is either 0 or a number divisible by 11.

e.g. 4832718

(Sum of digits at odd place) - (Sum of digits at even place)

$= (4+3+7+8) - (1+2+8) = 11 \Rightarrow$ which is divisible by 11.

10. Divisibility by 12 :- A number is divisible

by 12, if it is divisible by both 4 and 3.
e.g. 34632 ✓

11. Divisibility by 14 :- A number is divisible by 14, if it is divisible by 2 and 7 both.

12. Divisibility by 15 :- A number is divisible by 15, if it is divisible by both 3 and 5.

13. Divisibility by 16 :- A number is divisible by 16, if the number formed by the last 4 digits is divisible by 16.

e.g. 7957536 $\longrightarrow$ 7536 is divisible by 16.

14. Divisibility by 24 :- A given number is divisible by 24, if it is divisible by both 3 and 8.

15. Divisibility by 40 :- A given number is divisible by 40, if it is divisible by both 5 and 8.

16. Divisibility by 80 :- A given number is divisible by 80, if it is divisible by both 5 and 16.

* If a number is divisible by p as well as q , where p and q are co-primes, then the given number is divisible by pq .

If p and q are not co-primes, then the given numbers need not be divisible by pq , even when it is divisible by both p and q .

Ex. 36 is divisible by both 4 and 6, but ④ it is not divisible by $4 \times 6 = 24$, since 4 and 6 are not co-primes.

Multiplication by short-cut method :-

① Multiplication by distributive law :-

$$a(b+c) = ab + ac$$

$$a(b-c) = ab - ac$$

Ex. 567958×99999

$$= 567958 (100000 - 1)$$

$$= 567958 \times 10^5 - 567958 \times 1$$

$$= 567958 \text{ } \cancel{00000} - 567958$$

$$= 56795232042 \quad \underline{\text{Ans.}}$$

Ex. $978 \times 184 + 978 \times 816$

$$= 978 (184 + 816) = 978 \times 1000$$

$$= 978000 \quad \underline{\text{Ans.}}$$

Basic formulae

1. $(a+b)^2 = a^2 + b^2 + 2ab$

2. $(a-b)^2 = a^2 + b^2 - 2ab$

3. $(a+b)^2 - (a-b)^2 = 4ab$

4. $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$

5. $(a^2 - b^2) = (a-b)(a+b)$

6. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

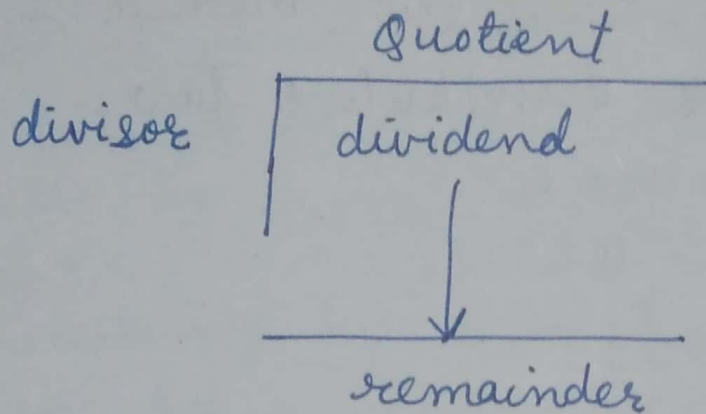
7. $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$

8. $a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$

9. $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

10. If $a+b+c=0$ then $a^3 + b^3 + c^3 = 3abc$

Division algorithm or Euclidean algorithm
If we divide a given number by another number, then



$$\begin{array}{r} 3 \overline{) 19} \\ \underline{- 18} \\ 1 \end{array}$$

- * $x^n - a^n$ is divisible by $x - a$ $\forall n$.
- * $x^n - a^n$ is divisible by $x + a$ $\forall$ even n .
- * $x^n + a^n$ is divisible by $x + a$ $\forall$ odd n .

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^2 + b^2 = \text{nothing}$$

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

Q. (1) On dividing $(9^3 - 7^3)$ by 2; remainder
✓ (a) 0 (b) 1 (c) 2 (d) 3

$$\begin{aligned} \text{Sol}^n :- 9^3 - 7^3 &= (9 - 7)(9^2 + 7^2 - 9 \cdot 7) \\ &= 2(9^2 + 7^2 - 9 \cdot 7) \end{aligned}$$

remainder is 0.

Ans.

Q. (2) $(9^4 - 7^4)$ is divisible by
(a) 2 (b) 4 (c) 8 (d) 16
✓ (e) all of the above.

$$\text{Sol}^n :- a^2 - b^2 = (a + b)(a - b)$$

$$= (9 + 7)(9 - 7) = 16 \cdot 2$$

Ans.

8 (3) On dividing 624 by 24, remainder is (5)
 (a) 0 (b) 1 (c) 2 (d) 3

Solⁿ:- $\frac{624}{24} = \frac{625-1}{24} = \frac{25^2-1^2}{24}$

$$25^2-1^2 = (25-1)(25+1) \\ = 24 \cdot 26$$

remainder = 0

Ans.

8 (4) On dividing 624 by 24, remainder is

(a) 0 (b) 1 (c) 2 (d) 3 Ans.

$$24 \Rightarrow 12, 8, 6, 4, 2$$

8 (5) Find the remainder when $3^4 - 2^4$ is divided by 5 :- (a) 3 (b) 2 (c) 1 (d) 0

Solⁿ:- $(3-2)(3+2) = 1 \cdot 5$

is divisible by 1 & 5

remainder = 0

Ans.

(6) $5^{12} - 2^{12}$ is divisible by

(a) 7 (b) 14 (c) 21 (d) all of these

Solⁿ:- $a^2 - b^2 = (a-b)(a+b)$

$$(5-2)(5+2) \\ 3 \cdot 7$$

(7) On dividing $(9^3 + 7^3)$ by 4, remainder is

(a) 0 (b) 1 (c) 2 (d) 3

$$9+7=16 \Rightarrow \text{divisible by } \begin{matrix} 16 \\ 8 \\ 4 \\ 2 \end{matrix}$$

remainder = 0

Ans.

Q ⑧ On dividing $4^9 + 3^9$ by 7,
remainder is
✓ (a) 0 (b) 1 (c) 2 (d) 3

Solⁿ:- $4 + 3 = 7$

remainder = 0

✓ Ans.

Q ⑨ On dividing $6^{17} + 2^{17}$ is divisible by
(a) 2 (b) 4 (c) 8 (d) all above

Solⁿ:- $6 + 2 = \frac{8}{4}{2}$

✓ Ans.

Q ⑩ * $\frac{11}{10} \rightarrow \text{remainder} = 1$ | $\frac{6}{5} = 1$
 $\frac{11}{10} \stackrel{100}{=} \text{remainder} (1)^{100} = 1$ | $\frac{7}{8} = -1 + 8 =$
 $\frac{14}{15} = -1 + 15 = 1$

Q ⑩ $68^{88} \div 67$ Find remainder $\frac{14^3}{15}$
 $\frac{68^{88}}{67} = \frac{68}{67} \stackrel{\text{remainder}}{=} 1^{88} = 1$

Q ⑪ $66^{99} \div 67$ Find remainder

Solⁿ:- $(-1)^{99} + 67 = -1 + 67 = 66$ ✓ Ans.

Q ⑫ $65^3 \div 67$

Solⁿ:- $(-2)^3 + 67 = -8 + 67 = 59$ ✓ Ans.

Q ⑬ $(18^{15} + 4) \div 17$

Solⁿ:- $\frac{18^{15} + 4}{17} = \frac{18^{15}}{17} + \frac{4}{17} = (1)^{15} + (-13)$
 $= 1 - 13 + 17 = 1 + 4 = 5$ ✓ Ans.

Progression :-

A succession of numbers formed and arranged in a definite order according to certain definite rule, is called a progression.

1. Arithmetic Progression (A.P.) :- If each term of a progression differs from its preceding term by a constant, then such a progression is called an arithmetic progression. This constant difference is called the common difference of the A.P.

An A.P. with first term a and common difference d is given by

$$a, a+d, a+2d, a+3d, \dots, a+nd, \dots$$

The n^{th} term of this A.P. is

$$T_n = a + (n-1)d$$

The sum of n terms of this A.P.

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} (\text{first term} + \text{last term})$$

Some important results :-

$$(i) \quad 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$(ii) \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(iii) \quad 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

2. Geometric Progression (G.P.) :-

A progression of numbers in which every term bears a constant ratio with its preceding term, is called a geometrical progression.

The constant ratio is called the common ratio of the G.P.

A G.P. with first term a and common ratio r is :

$$a, ar, ar^2, ar^3, \dots$$

In this G.P.

$$T_n = ar^{n-1}$$

Sum of the n terms

$$S_n = \frac{a(1-r^n)}{1-r}$$

Q. what could be the maximum value of Q in the following question?

$$5P9 + 3R7 + 2Q8 = 1114$$

Solⁿ we may analysis

the given eqⁿ as shown :

Clearly P

$$2 + P + R + Q = 11$$

①	②	
5	P	9
3	R	7
2	Q	8
11	1	4

So, the maximum value of Q can be

(11-2) i.e. 9 (when $P=0, R=0$) Ans. (7)

ex. Simplify

(i) 5793405×9999

Solⁿ:- $5793405 \times (10000 - 1)$

$= 57934050000 - 5793405$

$= 57928256595$ Ans.

(ii) 839478×625

Solⁿ:-

ex. Evaluate

(i) $986 \times 137 + 986 \times 863$

Solⁿ:- $986 (137 + 863)$

$= 986 (1000) = 986000$ Ans.

(ii) $983 \times 207 - 983 \times 107$

Solⁿ:- $983 \times (207 - 107)$

$= 983 \times 100 = 98300$ Ans.

Ex. Simplify :

(i) 1605×1605

Solⁿ:- $(1605)^2 = (1600 + 5)^2$
 $= 2560000 + 25 + 16000$
 $= 2576025$ Ans.

(ii) ~~$983 \times 207 = 983 \times 107$~~
 1398×1398

Solⁿ:- $(1398)^2 = (1400 - 2)^2$
 $= 1960000 + 4 - 5600$
 $= 1954404$ Ans.

Ex. Evaluate $313 \times 313 + 287 \times 287$

$\therefore a^2 + b^2 = \frac{1}{2} [(a+b)^2 + (a-b)^2]$
 $\therefore (313)^2 + (287)^2 = \frac{1}{2} [(313+287)^2 + (313-287)^2]$
 $= \frac{1}{2} [(600)^2 + (26)^2]$
 $= \frac{1}{2} [360000 + 676]$
 $= 180338$ Ans.

Ex. Find the unit's digit in the product

(8)

$$(2467)^{153} \times (341)^{72}$$

Solⁿ:- Clearly, unit's digit in the given product
= unit's digit in $7^{153} \times 1^{72}$

Now 7^4 gives unit digit 1.

$\therefore 7^{152}$ gives unit digit 1.

$\therefore 7^{153}$ gives unit digit $(1 \times 7) = 7$.

Also, 1^{72} gives unit digit 1.

Hence, unit's digit in the product

$$= 7 \times 1 = 7. \quad \text{Ans.}$$

Ex. Find the unit's digit in $(264)^{102} + (264)^{103}$

Solⁿ:- Required unit's digit

$$= \text{unit's digit in } 4^{102} + 4^{103}$$

Now, 4^2 gives unit digit 6

$$\therefore 4^{102} \quad \text{-----} \quad 6$$

$$4^{103} \quad \text{-----} \quad 6 \times 4 \text{ i.e. } 4$$

$$\text{Hence unit's digit in } (264)^{102} + (264)^{103} \\ = 6 + 4 = 0 \quad \text{Ans.}$$

Ex. Find the total number of prime factors in the expression $4^{11} \times 7^5 \times 11^2$

Solⁿ: - $4^{11} \times 7^5 \times 11^2$

$$= (2 \times 2)^{11} \times 7^5 \times 11^2$$

$$= 2^{11} \times 2^{11} \times 7^5 \times 11^2$$

$$= 2^{22} \times 7^5 \times 11^2$$

∴ Total no. of prime factors

$$= 22 + 5 + 2 = \text{Ans. } 29$$

Ex. what least value must be assigned to * so that the number 197*5462 is divisible by 9?

Solⁿ: - let missing digit be x.

Sum of digits be

$$1 + 9 + 7 + x + 5 + 4 + 6 + 2 = 34 + x$$

$$= 3 + 4 + x = 7 + x$$

$$\Rightarrow x = 2 \quad \text{Ans.}$$

Ex. which digits should come in place of * and \$ if the number 62684*\$ is divisible by both 8 and 5?

Solⁿ: - Since the given number is divisible by 5, so 0 or 5 must come in place of \$. But a number ending with 5 is never divisible by 8, so, 0 will replace \$.

Now, the number formed by the last three digits is $4 * 0$, which becomes divisible by 8 if $*$ is replaced by 4.

Hence, digits in place of $*$ and $\$$ are 4 and 0 resp. Ans.

Ex: What least number must be added to 3000 to obtain a number exactly divisible by 19?

Solⁿ:— On dividing 3000 by 19, 17 is remainder.

$\therefore$ Number to be added $= 19 - 17 = 2$ Ans.

Ex: What least number must be subtracted from 2000 to get a number exactly divisible by 17?

Solⁿ:— On dividing 2000 by 17, we get 11 as remainder.

$\therefore$ Required number to be subtracted $= 11$ Ans.

Ex: Find the number which is nearest to 3105 and is exactly divisible by 21.

Solⁿ:— On dividing 3105 by 21, we get 18 as remainder.

$\therefore$ Number to be added to 3105 $= (21 - 18) = 3$

$\therefore$ required number $= 3105 + 3 = 3108$ Ans.

Ex: Find the smallest number of 6 digits which is exactly divisible by 111.

Solⁿ:- Smallest number of 6 digits is 100000.
On dividing 100000 by 111, we get 100 as remainder.

$$\therefore \text{Number to be added} = 111 - 100 = 11$$

Hence, required number = 100011. Ans.

Ex. On dividing 15968 by a certain number, the quotient is 89 and the remainder is 37. Find the divisor.

Solⁿ:- $\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$

$$\Rightarrow x \overline{) 15968} \begin{matrix} 89 \\ \hline 37 \end{matrix}$$

$$\begin{array}{r} \text{quotient} \\ \text{Divisor } \overline{) \text{Dividend}} \\ \downarrow \\ \hline \text{remainder} \end{array}$$

$$3 \overline{) 13} = 4$$

$$\Rightarrow 15968 = 89 \times x + 37$$

$$\Rightarrow x = \frac{15968 - 37}{89} = 179 \quad \text{Ans.}$$

Ex. A number when divided by 342 gives a remainder 47. When the same number is divided by 19, what would be the remainder.

Solⁿ:- On dividing the given number by 342, let K be the quotient and 47 as remainder.

$$342 \overline{) x} \begin{matrix} K \\ \hline 47 \end{matrix}$$

$$\text{Then } x = 342K + 47$$

$$\Rightarrow x = 19 \times 18K + 19 \times 2 + 9 = 19(18K + 2)$$

(10)

The given number when divided by 19, gives $(18k+2)$ as quotient and 9 as remainder. Ans.

Ex: Find the remainder when 2^{31} is divided by 5.

Solⁿ:- $2^{10} = 1024$.

Unit digit of $2^{10} \times 2^{10} \times 2^{10}$ is $4 \times 4 \times 4$ is 4.

$\therefore$ unit digit of 2^{31} is 8 (4×2)

Now, 8 when divided by 5, gives 3 as remainder.

Hence, 2^{31} when divided by 5, gives 3 as remainder. Ans.

Ex: How many numbers are between 11 and 90 are divisible by 7?

Solⁿ:- The required numbers are

14, 21, ~~25~~ 28, 35, 42, 49, - - - - , 70, 77, 84.

This is an A.P. with $a=14$ and $d=7$
let it contains n terms.

Then $T_n = 84$

$\Rightarrow a + (n-1)d = 84$

$\Rightarrow 14 + (n-1) \cdot 7 = 84 \Rightarrow 7(n-1) = 70$

$\Rightarrow n-1 = 10$

$\Rightarrow n = 11$

$\therefore$ Required number of terms = 11

Ans.

Ex. Find the sum of all odd numbers upto 100

Solⁿ:- The given numbers are

1, 3, 5, 7, ..., 99.

This is an A.P. with $a = 1$, $d = 2$

let it contains n terms. Then

$$a + (n-1)d = 99$$

$$\Rightarrow 1 + (n-1)2 = 99$$

$$\Rightarrow 2(n-1) = 98$$

$$\Rightarrow n-1 = 49$$

$$\Rightarrow n = 50$$

$$\therefore \text{Required sum} = \frac{n}{2} [\text{first term} + \text{last term}]$$

$$= \frac{50}{2} (1 + 99) = 2500. \quad \underline{\text{Ans.}}$$

Ex. Find the sum of all 2 digit numbers divisible by 3.

Solⁿ:- All 2 digit numbers divisible by 3 are

12, ~~15~~ 15, 18, 21, ..., 99

This is an A.P. with $a = 12$, $d = 3$

let it contain n terms. Then

$$a + (n-1)d = 99$$

$$\Rightarrow 12 + (n-1) \cdot 3 = 99$$

$$\Rightarrow 3(n-1) = 87$$

$$\Rightarrow n-1 = 29$$

$$\Rightarrow n = 30$$

$$\therefore \text{Required sum} = \frac{n}{2} [\text{first term} + \text{last term}]$$
$$= \frac{30}{2} [12 + 99] = 1665 \quad \underline{\text{Ans}}$$

Ex. How many terms are there in

2, 4, 8, 16, ..., 1024 ?

Solⁿ:- Clearly 2, 4, 8, 16, ..., 1024 form a G.P. with $a = 2$, $r = \frac{4}{2} = 2$.

Let the number of terms be n . Then,

$$T_n = ar^{n-1}$$

$$\Rightarrow 1024 = 2 \cdot 2^{n-1}$$

$$\Rightarrow 2^{n-1} = 512 = 2^9$$

$$\Rightarrow n-1 = 9 \quad \Rightarrow n = 10 \quad \underline{\text{Ans.}}$$

Ex. $2 + 2^2 + 2^3 + \dots + 2^8 = ?$

Solⁿ:- Given series is a G.P. with $a = 2$, $r = 2$ and $n = 8$

$$\therefore \text{Sum} = \frac{a(r^n - 1)}{r - 1} = \frac{2(2^8 - 1)}{2 - 1}$$

$$= 2 \times 255 = 510 \quad \underline{\text{Ans.}}$$

————— X —————